Problem 6 – ProofCouncil

Abstract

An L-space knot is a knot in S^3 with a non-trivial Dehn surgery whose Heegaard Floer homology is as small as its first homology allows. We show that every Dehn surgery on an L-space knot has at most two non-trivial prime summands. The proof combines published results on reducible surgeries (L-space knots are fibered, by Ni; a reducible surgery on a hyperbolic knot with a positive L-space surgery has slope $2g-1$, by Hom–Lidman–Zufelt; the Cabling Conjecture holds for non-trivial non-hyperbolic knots, by Moser and Scharlemann; Gordon's formula for surgery on a cable; the Gordon–Luecke theorem on non-integral surgeries) with an elementary group-theoretic obstruction: if a p/q-surgery on a fibered knot of genus g has a non-trivial integer homology sphere summand, then $|p| \leq g$. The obstruction is proved by estimating the minimal number of generators of the commutator subgroup of the fundamental group in two ways, through the universal abelian cover and Grushko's theorem on free products on one side and through the fundamental group of the fiber surface on the other.

1 Introduction

Dehn surgery along a knot $K \subset S^3$ produces a closed oriented 3-manifold $S^3_r(K)$ for every slope $r \in \mathbb{Q} \cup \{\infty\}$ (notation is recalled in Section 2.1). By the Kneser–Milnor theorem, every closed oriented 3-manifold is a connected sum of prime manifolds, unique up to order, and a basic question in Dehn surgery theory is how complicated this decomposition can be for a surgery on a knot. For a non-trivial knot $K \subset S^3$, the Cabling Conjecture of González-Acuña and Short [3] predicts that $S^3_r(K)$ is reducible only when K is a cable knot and r is the slope of the cabling annulus (the unknot must be excluded, since its 0-surgery is $S^1 \times S^2$); in that case the surgery is a lens space connected-summed with a surgery on the companion knot (Gordon [5]), and the latter surgery is prime, because its slope is non-integral (Gordon–Luecke [6]) or because the companion is the unknot, so the conjecture predicts in particular that a surgery on a knot in S^3 never has three non-trivial prime summands. The conjecture is known for torus knots (Moser [15]) and for satellite knots (Scharlemann [21]), and it is open for hyperbolic knots. Without any hypothesis on the knot, Howie [11] showed that at most two prime summands of a surgery on a knot in S^3 can have non-zero first homology, so a third summand would have to be an integer homology sphere; see also [12].

This note answers the question for knots satisfying a strong Heegaard Floer restriction. An *L-space* is a rational homology sphere Y whose Heegaard Floer homology $\widehat{HF}(Y)$ has the

smallest possible dimension, namely $|H_1(Y;\mathbb{Z})|$; lens spaces are examples. A knot $K \subset S^3$ is an *L-space knot* if $S^3_r(K)$ is an L-space for some slope $r \neq \infty$. We allow r of either sign; some authors require $r > 0$, and we explain at the end of Section 2.5 and in Step 1 of Section 5 why the two conventions lead to the same theorem. These notions are recalled in Section 2.5. Torus knots are L-space knots, since suitable surgeries on them are lens spaces [15].

We count the prime summands of a manifold other than S^3 in its Kneser–Milnor decomposition, and we say that S^3 has none (Section 2.2); these are its *non-trivial prime summands*.

Theorem 1.1. *Let $K \subset S^3$ be an L-space knot. For every slope $r \in \mathbb{Q} \cup \{\infty\}$, the manifold $S^3_r(K)$ has at most two non-trivial prime summands.*

The L-space hypothesis enters the proof at exactly two points, each through a published theorem: L-space knots are fibered (Ni [16]), and a reducible surgery on a *hyperbolic* knot admitting a positive L-space surgery has slope exactly $2g - 1$, where g is the genus of the knot (Hom–Lidman–Zufelt [10, Theorem 1.3]). For non-trivial non-hyperbolic knots no Floer-theoretic input is needed, because the Cabling Conjecture is a theorem there. The argument does not identify the reducible surgery itself, and it does not prove the Cabling Conjecture for hyperbolic L-space knots; see Remark 6.2. The main ingredient we prove here is the following statement about fibered knots, established in Section 3 as Proposition 3.4.

Theorem 1.2. *Let $K \subset S^3$ be a fibered knot of genus $g \geq 1$, and let p/q be a slope with $p \neq 0$ and $|p| \geq 2g - 1$. Then $S^3_{p/q}(K)$ has at most two non-trivial prime summands.*

Strategy. Suppose that $M = S^3_{p/q}(K)$ has three non-trivial prime summands. By Howie's theorem one of them, call it Σ, has $H_1(\Sigma;\mathbb{Z}) = 0$, so $M \cong X \# \Sigma$ with Σ an integer homology sphere other than S^3 and $H_1(X;\mathbb{Z}) \cong H_1(M;\mathbb{Z}) \cong \mathbb{Z}/|p|$. Let $G = \pi_1(M)$ and let G' be its commutator subgroup. We compute the minimal number of generators $d(G')$ in two ways.

- *From above*, via the fiber. The group G is a quotient of the knot group $\pi_1(E_K)$, where E_K is the knot exterior, whose commutator subgroup is the fundamental group of the fiber surface, a free group of rank $2g$. An epimorphism maps commutator subgroups onto commutator subgroups, so $d(G') \leq 2g$ (Lemma 3.1).

- *From below*, via the universal abelian cover. The subgroup G' is the fundamental group of the $|p|$-fold cover $\widetilde{M} \to M$ with deck group $G/G' = H_1(M;\mathbb{Z})$. Because $\pi_1(\Sigma)$ is perfect, it maps trivially to the deck group, so the punctured copy of Σ inside M lifts to $|p|$ disjoint copies, and $\widetilde{M} \cong \widetilde{X} \# \Sigma^{\#|p|}$. Since $\pi_1(\Sigma)$ is non-trivial and perfect it is not cyclic, so it needs at least two generators, and Grushko's theorem gives $d(G') \geq 2|p|$ (Lemma 3.2).

Hence $|p| \leq g$. This is incompatible with $|p| \geq 2g - 1$ unless $g = 1$ and $|p| = 1$, an endpoint case that is excluded by a direct application of Grushko's theorem to the perfect group G.

Theorem 1.1 then follows by Thurston's trichotomy, according to which every non-trivial knot is a torus knot, a satellite knot or hyperbolic (Section 2.3). For a hyperbolic L-space

knot, Ni's theorem and the Hom–Lidman–Zufelt theorem put every reducible surgery exactly in the range of Theorem 1.2; for a torus or satellite knot the Cabling Conjecture, Gordon's formula for surgery on a cable, and the Gordon–Luecke theorem that non-integral surgeries are irreducible leave room for at most two summands.

Outline. Section 2 fixes conventions and recalls the standard facts we use, Section 3 proves the group-rank obstruction, Section 4 states the published theorems that are used as black boxes, Section 5 proves Theorem 1.1, and Section 6 discusses what the argument does and does not show.

A genus bound of the shape $|p| \leq g$ for surgeries with a non-trivial integer homology sphere summand is attributed to González-Acuña and Short [3, Proposition 1.4] in [1, Theorem 1.2], in the case where the complementary summand X is a lens space. The case needed here has X non-prime, so we prove Lemma 3.2 for an arbitrary complementary summand. Related statements about homology sphere summands of surgeries on L-space knots appear in [1, Section 4]. We make no claim of priority for Theorem 1.1; the aim of this note is a complete and self-contained argument modulo the cited theorems.

2 Conventions and background

2.1 Dehn surgery

Throughout, $K \subset S^3$ is a tame knot, νK is a closed tubular neighbourhood of K, and $E_K = S^3 \setminus \operatorname{int} \nu K$ is the *exterior* of K, a compact oriented 3-manifold with torus boundary. On $\partial E_K = \partial \nu K$ we fix the *meridian* μ, a simple closed curve bounding a disc in νK, and the *preferred longitude* λ, a simple closed curve isotopic to K in νK and null-homologous in E_K (it is the boundary of a Seifert surface for K). A *slope* is an isotopy class of unoriented essential simple closed curves on ∂E_K. The curve homologous to $p\mu + q\lambda$, with $\gcd(p, q) = 1$, determines the slope $p/q \in \mathbb{Q} \cup \{\infty\}$, where $\infty = 1/0$ is the meridional slope; for finite slopes we normalise $q > 0$, so that $p \in \mathbb{Z}$ is determined.

The *Dehn surgery* on K with slope r is the closed oriented 3-manifold

$$S^3_r(K) = E_K \cup_\varphi (S^1 \times D^2),$$

where the gluing homeomorphism $\varphi\colon \partial(S^1 \times D^2) \to \partial E_K$ sends the meridian $\{*\} \times \partial D^2$ of the solid torus to a curve of slope r, and the orientation extends that of $E_K \subset S^3$. Its homeomorphism type depends only on r. We use the following standard facts.

(S1) $S^3_\infty(K) = S^3$, since filling along the meridian undoes the removal of νK.

(S2) $H_1(E_K; \mathbb{Z}) \cong \mathbb{Z}$, generated by μ, and $\lambda = 0$ in $H_1(E_K; \mathbb{Z})$. Filling along $p\mu + q\lambda$ kills the class $p[\mu]$, so

$$H_1(S^3_{p/q}(K); \mathbb{Z}) \cong \mathbb{Z}/|p|\mathbb{Z},$$

which is to be read as $\mathbb{Z}$ when $p = 0$. In particular $S^3_{p/q}(K)$ is a rational homology sphere if and only if $p \neq 0$.

(S3) By van Kampen's theorem, $\pi_1(S^3_{p/q}(K)) \cong \pi_1(E_K)/\langle\langle \gamma_{p/q} \rangle\rangle$, the quotient by the normal closure of the element $\gamma_{p/q}$ of $\pi_1(E_K)$ represented by the filling curve of slope p/q (the elements of $\pi_1(E_K)$ represented by μ and λ are again denoted μ and λ). Applied to $p/q = \infty$ this gives $\pi_1(E_K)/\langle\langle \mu \rangle\rangle \cong \pi_1(S^3) = 1$: the knot group is normally generated by a meridian.

(S4) Let $\overline{K}$ be the mirror image of K. A reflection of S^3 carrying K to $\overline{K}$ preserves meridians and preferred longitudes but reverses the intersection pairing on the boundary torus, so it carries the slope r of K to the slope $-r$ of $\overline{K}$ and induces an orientation-reversing homeomorphism $S^3_r(K) \to S^3_{-r}(\overline{K})$; equivalently, there is an orientation-preserving identification $S^3_{-r}(\overline{K}) \cong -S^3_r(K)$, where $-M$ denotes M with the opposite orientation.

(S5) For the unknot U, $S^3_{p/q}(U)$ is a lens space with first homology of order $|p|$ (namely $L(p,q)$ up to orientation) when $|p| \geq 2$, is S^3 when $|p| = 1$, and is $S^1 \times S^2$ when $p = 0$. Here, for $p \geq 2$ (negative p differs only by orientation), $L(p,q)$ denotes the quotient of the unit sphere $S^3 \subset \mathbb{C}^2$ by the free action of $\mathbb{Z}/p\mathbb{Z}$ generated by $(z,w) \mapsto (\zeta z, \zeta^q w)$, $\zeta = e^{2\pi i/p}$; it has $H_1(L(p,q);\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$ and universal cover S^3. Orientation conventions for lens spaces play no role in this note, since only the number of prime summands is counted.

2.2 Prime decomposition

Let M be a closed connected oriented 3-manifold. M is *irreducible* if every embedded 2-sphere in M bounds a 3-ball, and *reducible* otherwise. M is *prime* if $M \cong M_1 \# M_2$ implies $M_1 \cong S^3$ or $M_2 \cong S^3$. An irreducible manifold is prime, and conversely a prime closed orientable 3-manifold is either irreducible or homeomorphic to $S^1 \times S^2$ [9, §1.1]. Thus for a rational homology sphere, prime and irreducible are the same thing, and non-prime is the same as reducible.

If a covering space of M is irreducible, so is M [9, Proposition 1.6]; by Alexander's theorem [9, §1.1], S^3 and $\mathbb{R}^3$ are irreducible. Hence lens spaces, which are covered by S^3, are irreducible, and so is any closed 3-manifold with universal cover $\mathbb{R}^3$.

The Kneser–Milnor theorem [9, Theorem 1.5] states that every closed oriented 3-manifold $M \ncong S^3$ is a connected sum $M \cong M_1 \# \cdots \# M_k$ of prime manifolds $M_i \ncong S^3$, and that the collection $\{M_i\}$ is unique up to order and orientation-preserving homeomorphism. We call $M_1, \ldots, M_k$ the *non-trivial prime summands* of M; by uniqueness their number k is an invariant of M, and we set $k = 0$ for $M \cong S^3$. Reversing the orientation of M reverses that of each M_i, so M and $-M$ have the same number of non-trivial prime summands. By van Kampen's theorem, for any connected sum of closed oriented 3-manifolds $N_1, \ldots, N_k$ (prime or not),

$$\pi_1(N_1 \# \cdots \# N_k) \cong \pi_1(N_1) * \cdots * \pi_1(N_k), \qquad H_1(N_1 \# \cdots \# N_k;\mathbb{Z}) \cong \bigoplus_{i=1}^{k} H_1(N_i;\mathbb{Z}), \quad (1)$$

since the connected sum is formed by gluing punctured copies of the N_i along 2-spheres, and removing a ball from a 3-manifold does not change its fundamental group.

A closed connected oriented 3-manifold Σ with $H_1(\Sigma;\mathbb{Z}) = 0$ is called an *integer homology sphere*; by Poincaré duality and the universal coefficient theorem, $H_*(\Sigma;\mathbb{Z}) \cong H_*(S^3;\mathbb{Z})$.

2.3 Torus, satellite and hyperbolic knots; cables

A non-trivial knot $K \subset S^3$ is a *torus knot* if it can be isotoped onto the boundary of a standardly embedded solid torus; it is a *satellite knot* if E_K contains an incompressible torus (an embedded torus whose fundamental group injects into $\pi_1(E_K)$) that is not parallel to ∂E_K; and it is *hyperbolic* if $S^3 \setminus K$ admits a complete hyperbolic metric of finite volume. By Thurston's hyperbolization theorem, every non-trivial knot is of exactly one of these three types [22, Corollary 2.5]. Composite knots are satellite knots: if $K = K_1 \# K_2$ with both summands non-trivial, the boundary of a regular neighbourhood of the union of K with a ball containing the K_2 summand is a torus in E_K, knotted like K_1, which is incompressible and not boundary-parallel (the "swallow–follow" torus).

Let $J \subset S^3$ be a knot, with meridian μ_J and preferred longitude λ_J on $\partial \nu J$, and let $w \geq 2$ and t be coprime integers (so $t \neq 0$). The (w,t)-*cable* of J is the knot

$$C_{w,t}(J) \subset \partial \nu J \subset S^3$$

given by a simple closed curve on $\partial \nu J$ homologous to $t\mu_J + w\lambda_J$ there. Its image in $H_1(\nu J;\mathbb{Z}) \cong \mathbb{Z}$ is w times the generator, so w is the *winding number* of the cable. If J is the unknot, $C_{w,t}(J)$ is the torus knot $T_{w,t}$; if J is non-trivial, $C_{w,t}(J)$ is a satellite knot with companion J. We call wt the *cabling slope* of $C_{w,t}(J)$: it is the framing of the cable induced by the torus $\partial \nu J$, and the annulus $\partial \nu J \setminus \operatorname{int} \nu C_{w,t}(J)$ has both boundary curves of slope wt on $\partial \nu C_{w,t}(J)$. (Several indexing conventions for cables are in use; we fix the one above, in which the first index is the winding number.)

2.4 Fibered knots

A knot K is *fibered* if E_K is the total space of a fiber bundle over S^1 whose fibers are Seifert surfaces for K, that is, compact connected oriented surfaces $F \subset E_K$ with $\partial F = \lambda$. The fiber then realises the minimal genus of a Seifert surface for K, so its genus is the *genus* $g = g(K)$ of the knot, and $g \geq 1$ when K is non-trivial. Since F is a compact connected orientable surface of genus g with one boundary component, it deformation retracts onto a wedge of $2g$ circles, and $\pi_1(F)$ is a free group F_{2g} of rank $2g$. The long exact homotopy sequence of the bundle gives a short exact sequence

$$1 \longrightarrow \pi_1(F) \longrightarrow \pi_1(E_K) \longrightarrow \pi_1(S^1) = \mathbb{Z} \longrightarrow 1, \tag{2}$$

because $\pi_2(S^1) = 0$ and F is connected. The meridian μ meets each fiber transversally in a single point, so the map to $\mathbb{Z}$ in (2) sends μ to a generator.

2.5 Heegaard Floer homology and L-spaces

Heegaard Floer homology [18] assigns to a closed oriented 3-manifold Y a finite-dimensional vector space $\widehat{HF}(Y)$ over $\mathbb{F}_2$ (we use $\mathbb{F}_2$ coefficients throughout), invariant up to homeomorphism. If Y is a rational homology sphere, then $\dim_{\mathbb{F}_2} \widehat{HF}(Y) \geq |H_1(Y;\mathbb{Z})|$: the group splits as a direct sum indexed by the spin^c structures of Y, of which there are $|H^2(Y;\mathbb{Z})| = |H_1(Y;\mathbb{Z})|$, and each summand has Euler characteristic ± 1 [19, Proposition 5.1]. We call Y an *L-space* if equality holds. (The original definition [20, Definition 1.1] requires $\widehat{HF}(Y;\mathbb{Z})$ to be a free abelian group of rank $|H_1(Y;\mathbb{Z})|$; the $\mathbb{F}_2$-version above is the one commonly used in the Heegaard Floer literature, and it is the one used in this note.) Lens spaces, including S^3, are L-spaces [19, Proposition 3.1], which explains the name. The group $\widehat{HF}(-Y)$ is the dual of $\widehat{HF}(Y)$ [19, Proposition 2.5], so Y is an L-space if and only if $-Y$ is.

A knot $K \subset S^3$ is an *L-space knot* if $S^3_r(K)$ is an L-space for some $r \in \mathbb{Q}$; by (S2) such an r is automatically non-zero. We say that K admits a *positive* L-space surgery if this holds for some $r > 0$. By (S4) and the invariance of the L-space condition under orientation reversal, K is an L-space knot if and only if K or its mirror image admits a positive L-space surgery. The unknot is an L-space knot by (S5), and so is every torus knot, since $(wt \pm 1)$-surgery on $T_{w,t}$ is a lens space [15].

2.6 Group-theoretic tools

For a finitely generated group G, let $d(G)$ denote the minimal number of generators of G, and let $G' = [G, G]$ denote its commutator subgroup. If $G \to Q$ is an epimorphism, then $d(Q) \leq d(G)$. Grushko's theorem [7] states that for finitely generated groups A and B,

$$d(A * B) = d(A) + d(B). \tag{3}$$

We use the following consequence of the Poincaré conjecture, proved by Perelman (see [14] for a detailed exposition).

Lemma 2.1. *Let Σ be an integer homology sphere with $\Sigma \ncong S^3$. Then $\pi_1(\Sigma)$ is a non-trivial perfect group, and $d(\pi_1(\Sigma)) \geq 2$.*

Proof. If $\pi_1(\Sigma)$ were trivial, then Σ would be a simply connected closed 3-manifold, hence homeomorphic to S^3 by the Poincaré conjecture. Its abelianization is $H_1(\Sigma;\mathbb{Z}) = 0$, so $\pi_1(\Sigma)$ is perfect. A non-trivial cyclic group has non-trivial abelianization, so $\pi_1(\Sigma)$ is not cyclic, that is, $d(\pi_1(\Sigma)) \geq 2$. □

A group G is *normally generated* by a subset S if the normal closure $\langle\langle S \rangle\rangle$ is all of G; the *weight* of G is the minimal size of such an S. If G is normally generated by one element and $G \to Q$ is an epimorphism, then Q is normally generated by the image of that element. By (S3), the fundamental group of every Dehn surgery on a knot in S^3 is normally generated by the image of a meridian, that is, has weight at most one. The group-theoretic input that limits the number of summands with non-zero homology is the following theorem of Howie, previously known as the Scott–Wiegold conjecture.

Theorem 2.2 (Howie [11]). *Let $a, b, c \geq 2$. The free product $(\mathbb{Z}/a\mathbb{Z}) * (\mathbb{Z}/b\mathbb{Z}) * (\mathbb{Z}/c\mathbb{Z})$ is not normally generated by a single element.*

3 The group-rank obstruction

In this section $K \subset S^3$ is a fibered knot of genus g, with exterior E_K and fiber F as in Section 2.4. No Floer-theoretic hypothesis is used here.

Lemma 3.1 (Upper bound from the fiber). *Let $K \subset S^3$ be a fibered knot of genus g, and let $M = S^3_r(K)$ for any slope $r \in \mathbb{Q} \cup \{\infty\}$. Then*

$$d\big(\pi_1(M)'\big) \leq 2g.$$

Proof. Put $\Pi = \pi_1(E_K)$ and $G = \pi_1(M)$. We first identify Π' with the fiber group. Let $\phi\colon \Pi \to \mathbb{Z}$ be the map in (2), so that $\ker\phi = \pi_1(F) \cong F_{2g}$. Since $\mathbb{Z}$ is abelian, ϕ factors as $\phi = \psi \circ \alpha$, where $\alpha\colon \Pi \to \Pi/\Pi' = H_1(E_K;\mathbb{Z}) \cong \mathbb{Z}$ is the abelianization and $\psi\colon \mathbb{Z} \to \mathbb{Z}$ is a homomorphism. Both α and ϕ send the meridian μ to a generator (by (S2) and Section 2.4, respectively), so ψ is onto, hence $\psi = \pm\mathrm{id}$ and $\ker\phi = \ker\alpha = \Pi'$. Thus $\Pi' = \pi_1(F) \cong F_{2g}$.

By (S3), Dehn filling induces an epimorphism $f\colon \Pi \to G$. For any epimorphism of groups, $f(\Pi') = G'$: the inclusion $f(\Pi') \subseteq G'$ is clear, and conversely every commutator $[x, y]$ in G equals $f([\tilde{x}, \tilde{y}])$ for lifts $\tilde{x}, \tilde{y}$ of x, y. Hence G' is a quotient of F_{2g}, and $d(G') \leq 2g$. □

The next lemma is the heart of the argument. The idea is that $\pi_1(M)'$ is the fundamental group of the universal abelian cover of M, and that in this cover a homology sphere summand of M is replicated once for each sheet, each replica contributing at least two generators.

Lemma 3.2 (Lower bound from the universal abelian cover). *Let M be a closed connected oriented 3-manifold with $H_1(M;\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ for some $n \geq 1$, and suppose that*

$$M \cong X \# \Sigma, \qquad H_1(\Sigma;\mathbb{Z}) = 0, \qquad \Sigma \ncong S^3,$$

for some closed oriented 3-manifold X. Then

$$d\big(\pi_1(M)'\big) \geq 2n.$$

In particular, if $M = S^3_{p/q}(K)$ with $p \neq 0$ for a fibered knot K of genus g, then a summand Σ as above can occur only if

$$|p| \leq g. \tag{4}$$

Proof. Choose closed balls $B_X \subset X$ and $B_\Sigma \subset \Sigma$ and write $M = X^\circ \cup_S \Sigma^\circ$, where $X^\circ = X \setminus \operatorname{int} B_X$ and $\Sigma^\circ = \Sigma \setminus \operatorname{int} B_\Sigma$ are the summands with the interior of a ball removed and S is the common boundary 2-sphere. By (1), $H_1(M;\mathbb{Z}) \cong H_1(X;\mathbb{Z}) \oplus H_1(\Sigma;\mathbb{Z}) = H_1(X;\mathbb{Z})$, so $H_1(X;\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$ and the inclusion $X^\circ \hookrightarrow M$ induces an isomorphism $H_1(X^\circ;\mathbb{Z}) \to H_1(M;\mathbb{Z})$.

Let $G = \pi_1(M)$ and let $\rho\colon \widetilde{M} \to M$ be the connected covering space corresponding to the normal subgroup G' (the *universal abelian cover* of M). It is a regular covering with deck group $G/G' \cong H_1(M;\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$, so it has n sheets, and $\pi_1(\widetilde{M}) \cong G'$. We use the following standard fact about a regular covering with deck group Q and a connected codimension-zero submanifold $A \subseteq M$ [8, §1.3]: if Q_A is the image of the composite $\pi_1(A) \to \pi_1(M) \to Q$, then $\rho^{-1}(A)$ has exactly $[Q : Q_A]$ components, and each component is the connected covering of A associated with the kernel of that composite.

We apply this to the two pieces of M. For $A = X^\circ$, the composite $\pi_1(X^\circ) \to G \to G/G' \cong H_1(M;\mathbb{Z})$ is the abelianization of $\pi_1(X^\circ)$ followed by the isomorphism $H_1(X^\circ;\mathbb{Z}) \to H_1(M;\mathbb{Z})$ noted above. So it is onto, and its kernel is $\pi_1(X^\circ)' = \pi_1(X)'$. Hence $\rho^{-1}(X^\circ)$ is connected, and it is the covering of X° associated with $\pi_1(X)'$. If $\widetilde{X} \to X$ denotes the connected n-sheeted covering associated with $\pi_1(X)'$, this says that

$$\rho^{-1}(X^\circ) \cong \widetilde{X}^\circ := \widetilde{X} \setminus (\text{the } n \text{ lifts of } \operatorname{int} B_X).$$

For $A = \Sigma^\circ$, the group $\pi_1(\Sigma^\circ) = \pi_1(\Sigma)$ is perfect by Lemma 2.1, so it maps trivially to the abelian deck group G/G'. Hence $\rho^{-1}(\Sigma^\circ)$ consists of n components, each mapped homeomorphically onto Σ°.

The n boundary spheres of $\widetilde{X}^\circ$ are the lifts of S, and each of them is glued to one copy of Σ°. Since $\widetilde{X}^\circ$ is connected, this exhibits

$$\widetilde{M} \cong \widetilde{X} \# \underbrace{\Sigma \# \cdots \# \Sigma}_{n \text{ copies}}. \tag{5}$$

All fundamental groups in sight are finitely generated, being fundamental groups of compact manifolds. By (1), Grushko's theorem (3) and Lemma 2.1,

$$d(G') = d\big(\pi_1(\widetilde{M})\big) = d\big(\pi_1(\widetilde{X})\big) + n\, d\big(\pi_1(\Sigma)\big) \geq 2n.$$

Finally, let $M = S^3_{p/q}(K)$ with $p \neq 0$ and K fibered of genus g. By (S2), $H_1(M;\mathbb{Z}) \cong \mathbb{Z}/|p|\mathbb{Z}$, so the first part applies with $n = |p|$ and gives $d(G') \geq 2|p|$, while Lemma 3.1 gives $d(G') \leq 2g$. Hence $|p| \leq g$. □

Howie's theorem (Theorem 2.2) limits the summands with non-zero homology; this is what produces the homology sphere summand to which Lemma 3.2 applies.

Lemma 3.3 (Consequence of Howie's theorem [11])**.** *Let $K \subset S^3$ be any knot and let p/q be a slope with $p \neq 0$. Then at most two non-trivial prime summands of $S^3_{p/q}(K)$ have non-zero first integral homology.*

Proof. Set $M = S^3_{p/q}(K)$ and let $M_1 \# \cdots \# M_k$ be its decomposition into non-trivial prime summands. By (1) and (S2),

$$\bigoplus_{i=1}^{k} H_1(M_i;\mathbb{Z}) \cong H_1(M;\mathbb{Z}) \cong \mathbb{Z}/|p|\mathbb{Z},$$

so each $H_1(M_i;\mathbb{Z})$ is a direct summand of a finite cyclic group and hence is finite cyclic. Suppose that three of them are non-zero, of orders $a, b, c \geq 2$. By (1), $\pi_1(M) \cong \pi_1(M_1) * \cdots * \pi_1(M_k)$. Abelianizing those three free factors and killing all the others defines an epimorphism

$$\pi_1(M) \twoheadrightarrow (\mathbb{Z}/a\mathbb{Z}) * (\mathbb{Z}/b\mathbb{Z}) * (\mathbb{Z}/c\mathbb{Z}).$$

By (S3), $\pi_1(M)$ is normally generated by one element, and this property passes to the quotient (Section 2.6). This contradicts Theorem 2.2. □

Proposition 3.4. *Let $K \subset S^3$ be a fibered knot of genus $g \geq 1$. If $p \neq 0$ and $|p| \geq 2g-1$, then $S^3_{p/q}(K)$ has at most two non-trivial prime summands.*

Proof. Suppose, for a contradiction, that $M = S^3_{p/q}(K) \cong M_1 \# \cdots \# M_k$ with $k \geq 3$ and each M_i a prime manifold other than S^3. By Lemma 3.3, at most two of the M_i have non-zero first homology, so after renumbering $H_1(M_k;\mathbb{Z}) = 0$; thus M_k is an integer homology sphere (Section 2.2) with $M_k \ncong S^3$. Apply Lemma 3.2 with $X = M_1 \# \cdots \# M_{k-1}$ and $\Sigma = M_k$: since $p \neq 0$, it gives $|p| \leq g$, and together with the hypothesis,

$$2g - 1 \leq |p| \leq g.$$

This forces $g \leq 1$, so $g = 1$ and $|p| = 1$.

It remains to rule out this endpoint. If $|p| = 1$, then $H_1(M;\mathbb{Z}) = 0$ by (S2), so by (1) every M_i is an integer homology sphere other than S^3. Put $G = \pi_1(M)$. Then G is perfect, so $G' = G$, and (1), Grushko's theorem (3) and Lemma 2.1 give

$$d(G') = d(G) = \sum_{i=1}^{k} d\big(\pi_1(M_i)\big) \geq 2k \geq 6.$$

But Lemma 3.1 gives $d(G') \leq 2g = 2$, a contradiction. □

Proposition 3.4 is Theorem 1.2. Note that it makes no assumption on the knot beyond fiberedness. The proof actually excludes a homology sphere summand whenever $|p| > g$, by (4). The threshold $|p| \geq 2g-1$ in the statement is the one that the Hom–Lidman–Zufelt theorem will supply for hyperbolic L-space knots; for $g = 1$ it admits the value $|p| = 1$, which is why the endpoint required the separate argument above.

Remark 3.5. Lemma 3.2 has a purely group-theoretic core: if $G = A * H$ with H perfect and $A/A' \cong \mathbb{Z}/n\mathbb{Z}$, then $G' \cong A' * H * \cdots * H$ with n copies of H. This can also be derived from the Kurosh subgroup theorem [13]; the covering space argument above makes the multiplicity n visible directly.

Example 3.6. Lemma 3.1 is sharp: for a torus knot, the cabling surgery already exhausts the bound $d(G') \leq 2g$ with a free group, leaving no room for a homology sphere summand. Let $T_{a,b}$ be a torus knot with $a, b \geq 2$ coprime, so that $2g(T_{a,b}) = (a-1)(b-1)$, and recall that $S^3_{ab}(T_{a,b}) \cong L(a,b) \# L(b,a)$ by Moser's theorem [15]. The fundamental group is $G = (\mathbb{Z}/a\mathbb{Z}) * (\mathbb{Z}/b\mathbb{Z})$, and the kernel G' of its abelianization onto $\mathbb{Z}/ab\mathbb{Z}$ is a free group of

rank $(a-1)(b-1) = 2g$: indeed, in the ab-fold universal abelian cover the punctured $L(a,b)$ lifts to b copies of S^3 with a balls removed, the punctured $L(b,a)$ lifts to a copies of S^3 with b balls removed, and these are glued along the ab lifts of the separating sphere; the resulting manifold is a connected sum of $ab-(a+b)+1 = (a-1)(b-1)$ copies of $S^1 \times S^2$. This is consistent with (4), since $ab > g$.

4 Inputs from Heegaard Floer theory and Dehn surgery

We now state the published theorems used in the proof of Theorem 1.1. None of them is reproved here.

Theorem 4.1 (Ni [16, Corollary 1.3]). *Every L-space knot in S^3 is fibered.*

Ni's corollary explicitly allows a rational L-space surgery slope, and its proof uses the structure theorem of Ozsváth and Szabó [20] for the knot Floer homology (the knot-level refinement of $\widehat{HF}$) of L-space knots; see also Ni's erratum [17], which repairs part of the proof of the fiberedness detection theorem without affecting the statement, and Ghiggini [4] for the genus-one case.

Theorem 4.2 (Hom–Lidman–Zufelt [10, Theorem 1.3]). *Let $K \subset S^3$ be a hyperbolic knot that admits a positive L-space surgery. If $S^3_r(K)$ is reducible, then $r = 2g(K)-1$.*

Two features of this statement matter for us. First, the reducible surgery itself is not assumed to be an L-space; the theorem constrains every reducible slope of K. Second, the hyperbolicity hypothesis cannot be dropped: the right-handed trefoil $T_{2,3}$ has genus 1 and admits a positive L-space surgery (its 5- and 7-surgeries are lens spaces [15]), but its reducible surgery $S^3_6(T_{2,3}) \cong L(2,3)\#L(3,2)$ has slope $6 \neq 2g-1$. This is why the proof of Theorem 1.1 treats non-hyperbolic knots separately, using the following known cases of the Cabling Conjecture.

Theorem 4.3 (Moser [15], Scharlemann [21, Corollaries 4.4–4.5]). *Let $K \subset S^3$ be a non-trivial knot that is not hyperbolic, that is, a torus knot or a satellite knot. If $S^3_r(K)$ is reducible, then $K = C_{w,t}(J)$ is a cable, with $w \geq 2$ and $\gcd(w,t) = 1$, of some knot J, and $r = wt$ is the cabling slope.*

For torus knots this is Moser's classification of surgeries on $T_{w,t} = C_{w,t}(U)$: only the slope wt yields a reducible manifold. For satellite knots it is Scharlemann's theorem; Corollary 4.4 of [21] treats a knot in a solid torus that is not contained in a ball, and Corollary 4.5 applies it to satellite knots in S^3. Since composite knots are satellite knots (Section 2.3), they are covered. The reduction of the Cabling Conjecture to hyperbolic knots along these lines is standard and is recorded, for instance, in the introduction of [10].

Theorem 4.4 (Gordon [5, Corollary 7.3]). *Let $J \subset S^3$ be a knot and let $w \geq 2$, t be coprime integers. Then, up to the orientation of the lens space,*

$$S^3_{wt}(C_{w,t}(J)) \cong L(w,t)\#S^3_{t/w}(J). \tag{6}$$

Formula (6) is consistent with (S2): both sides have first homology of order $|wt| = w \cdot |t|$, since $\gcd(w,t) = 1$. Geometrically, the cabling annulus together with two meridian discs of the filling solid torus forms a separating sphere exhibiting the decomposition (6): on one side it cuts off a punctured $S^3_{t/w}(J)$, and on the other a punctured lens space of order w, obtained from the solid torus νJ by attaching a 2-handle along a curve of winding number w. This sphere is a reducing sphere when both factors are non-trivial, but not always: for J the unknot and $|t| = 1$ the cable $C_{w,t}(J)$ is itself the unknot and $S^3_{t/w}(J) \cong S^3$, so the sphere bounds a ball.

Theorem 4.5 (Gordon–Luecke [6])**.** *If $J \subset S^3$ is a non-trivial knot and $r \in \mathbb{Q}$ is not an integer, then $S^3_r(J)$ is irreducible.*

5 Proof of Theorem 1.1

Let $K \subset S^3$ be an L-space knot and $r \in \mathbb{Q} \cup \{\infty\}$. We must show that $S^3_r(K)$ has at most two non-trivial prime summands.

Step 1: reduction to a positive L-space surgery. By the last paragraph of Section 2.5, either K or its mirror image $\overline{K}$ admits a positive L-space surgery. By (S4), $S^3_{-r}(\overline{K}) \cong -S^3_r(K)$, and by Section 2.2 a manifold and its orientation reversal have the same number of non-trivial prime summands. Mirroring also preserves the genus and the type (torus, satellite or hyperbolic) of a knot. Replacing (K, r) by $(\overline{K}, -r)$ if necessary, we may therefore assume that K admits a positive L-space surgery.

Step 2: the unknot. If K is the unknot, then by (S5) $S^3_r(K)$ is a lens space, S^3 or $S^1 \times S^2$. Lens spaces are irreducible and $S^1 \times S^2$ is prime (Section 2.2), so there is at most one non-trivial prime summand. Henceforth K is non-trivial; by Theorem 4.1 it is fibered, of genus $g \geq 1$.

Step 3: the slopes ∞ and 0. By (S1), $S^3_\infty(K) = S^3$ has no non-trivial prime summand. For the slope 0, the filling solid torus has meridian $\lambda = \partial F$, where F is a fiber; the fibration of E_K restricts on ∂E_K to the product fibration by longitudes, so the solid torus is a union of discs capping off the fibers, and $S^3_0(K)$ is a bundle over S^1 with fiber the closed surface $\widehat{F} = F \cup D^2$ of genus $g \geq 1$. The universal cover of $\widehat{F}$ is $\mathbb{R}^2$, so the pull-back of the bundle to the universal cover $\mathbb{R}$ of S^1 is $\widehat{F} \times \mathbb{R}$, and the universal cover of $S^3_0(K)$ is $\mathbb{R}^3$. Hence $S^3_0(K)$ is irreducible (Section 2.2; [9, Proposition 1.6]), so it is prime and has exactly one non-trivial prime summand (it is not S^3, having $H_1 \cong \mathbb{Z}$).

Step 4: finite non-zero slopes. Let $r = p/q$ with $p \neq 0$. If $S^3_r(K)$ is prime, it has at most one non-trivial prime summand. Otherwise it is reducible, and we distinguish the type of K (Section 2.3).

Case 4a: K is hyperbolic. By Step 1, K admits a positive L-space surgery, so Theorem 4.2 gives $r = 2g - 1$, that is, $p = 2g - 1 \geq 1$ and $q = 1$. Since K is fibered of genus $g \geq 1$, Proposition 3.4 applies and shows that $S^3_r(K)$ has at most two non-trivial prime summands.

Case 4b: K is a torus knot or a satellite knot. By Theorem 4.3, $K = C_{w,t}(J)$ with $w \geq 2$, $\gcd(w,t) = 1$, and $r = wt$. By Theorem 4.4, $S^3_r(K) \cong L(w,t) \# S^3_{t/w}(J)$. The first factor is a lens space with $|H_1| = w \geq 2$, hence a single non-trivial prime summand. The slope t/w is not an integer, because $w \geq 2$ and $\gcd(w,t) = 1$. If J is non-trivial, $S^3_{t/w}(J)$ is irreducible by Theorem 4.5, hence prime, and contributes at most one non-trivial prime summand. If J is the unknot, then $t \neq 0$ (as $\gcd(w,0) = w \geq 2$), so by (S5) $S^3_{t/w}(J)$ is a lens space or S^3, and again contributes at most one. By uniqueness of the prime decomposition, $S^3_r(K)$ has at most two non-trivial prime summands.

This completes the proof of Theorem 1.1. □

6 Comments

Remark 6.1 (Where the L-space hypothesis is used). The proof uses the L-space hypothesis only through Theorems 4.1 and 4.2, in Steps 2 and 4a (Step 1 merely arranges the sign of the L-space slope that Theorem 4.2 requires). Consequently the same argument proves the following statement, of which Theorem 1.1 is a special case: if K is a non-trivial fibered knot that is either non-hyperbolic, or hyperbolic with every reducible slope p/q satisfying $|p| \geq 2g(K) - 1$, then every surgery on K has at most two non-trivial prime summands. (Fiberedness is used in Step 3 for the slope 0 and in Step 4a through Proposition 3.4.)

Remark 6.2 (What is not proved). The argument does not establish the Cabling Conjecture for hyperbolic L-space knots. In Case 4a it handles a hypothetical reducible surgery at slope $2g - 1$ directly, by excluding the integer homology sphere summand that a third summand would force, without attempting to identify that summand by Heegaard Floer methods. It does not exclude a reducible surgery $S^3_{2g-1}(K)$ on a hyperbolic L-space knot with exactly two non-trivial prime summands, both with non-zero first homology. For an arbitrary knot in S^3, a surgery has at most three non-trivial prime summands, and three summands would have to be two lens spaces of coprime orders and an integer homology sphere; this combines results of Sayari, Valdez-Sánchez and Howie, see the introduction of [12]. Proposition 3.4 excludes the last configuration in the range $|p| \geq 2g - 1$ for fibered knots without using this classification.

Remark 6.3 (The ambient manifold). The hypothesis that K lies in S^3 is used throughout: in (S2) and (S3), in Ni's and Hom–Lidman–Zufelt's theorems, and in the Cabling Conjecture. For knots in other L-spaces, a surgery can have three non-trivial prime summands: a regular fiber of the Seifert fibration of the Poincaré homology sphere (itself an L-space) has a surgery homeomorphic to $L(2,1) \# L(3,2) \# L(5,4)$ [2].

References

[1] H. Bodish and R. DeYeso III, *Obstructing reducible surgeries: slice genus and thickness bounds*, arXiv:2209.01672 (2022).

[2] J. Caudell, *Three lens space summands from the Poincaré homology sphere*, arXiv:2101.01256 (2021).

[3] F. González-Acuña and H. Short, *Knot surgery and primeness*, Math. Proc. Cambridge Philos. Soc. **99** (1986), no. 1, 89–102.

[4] P. Ghiggini, *Knot Floer homology detects genus-one fibred knots*, Amer. J. Math. **130** (2008), no. 5, 1151–1169.

[5] C. McA. Gordon, *Dehn surgery and satellite knots*, Trans. Amer. Math. Soc. **275** (1983), no. 2, 687–708.

[6] C. McA. Gordon and J. Luecke, *Only integral Dehn surgeries can yield reducible manifolds*, Math. Proc. Cambridge Philos. Soc. **102** (1987), no. 1, 97–101.

[7] I. A. Grushko, *On the bases of a free product of groups*, Mat. Sbornik (N.S.) **8(50)** (1940), no. 1, 169–182 (Russian).

[8] A. Hatcher, *Algebraic topology*, Cambridge University Press, 2002.

[9] A. Hatcher, *Notes on basic* 3*-manifold topology*, unpublished notes, revised 2000.

[10] J. Hom, T. Lidman and N. Zufelt, *Reducible surgeries and Heegaard Floer homology*, Math. Res. Lett. **22** (2015), no. 3, 763–788; arXiv:1307.5317.

[11] J. Howie, *A proof of the Scott–Wiegold conjecture on free products of cyclic groups*, J. Pure Appl. Algebra **173** (2002), no. 2, 167–176.

[12] J. Howie, *Can Dehn surgery yield three connected summands?*, Groups Geom. Dyn. **4** (2010), no. 4, 785–797; arXiv:0908.2697.

[13] A. Kurosch, *Die Untergruppen der freien Produkte von beliebigen Gruppen*, Math. Ann. **109** (1934), 647–660.

[14] J. Morgan and G. Tian, *Ricci flow and the Poincaré conjecture*, Clay Mathematics Monographs **3**, Amer. Math. Soc., 2007.

[15] L. Moser, *Elementary surgery along a torus knot*, Pacific J. Math. **38** (1971), no. 3, 737–745.

[16] Y. Ni, *Knot Floer homology detects fibred knots*, Invent. Math. **170** (2007), no. 3, 577–608.

[17] Y. Ni, *Erratum: Knot Floer homology detects fibred knots*, Invent. Math. **177** (2009), no. 1, 235–238; arXiv:0808.0940.

[18] P. Ozsváth and Z. Szabó, *Holomorphic disks and topological invariants for closed three-manifolds*, Ann. of Math. (2) **159** (2004), no. 3, 1027–1158.

[19] P. Ozsváth and Z. Szabó, *Holomorphic disks and three-manifold invariants: properties and applications*, Ann. of Math. (2) **159** (2004), no. 3, 1159–1245.

[20] P. Ozsváth and Z. Szabó, *On knot Floer homology and lens space surgeries*, Topology **44** (2005), no. 6, 1281–1300.

[21] M. Scharlemann, *Producing reducible* 3*-manifolds by surgery on a knot*, Topology **29** (1990), no. 4, 481–500.

[22] W. P. Thurston, *Three dimensional manifolds, Kleinian groups and hyperbolic geometry*, Bull. Amer. Math. Soc. (N.S.) **6** (1982), no. 3, 357–381.